

1. TRISO particle layers, from the inside out:

- 1) Fuel Kernel - "does the fission stuff" produces heat for conversion to electricity or direct use (process heat)
- 2) Buffer layer - Acts as a sacrificial layer to accommodate fuel kernel swelling, absorb fission recoils, allow CO and fission product gases to expand
- 3) Inner Pyrolytic Carbon - Protect SiC from attack (fission products and CO), puts SiC into compression to increase its strength (IPyC)
- 4) Silicon Carbide - Pressure boundary - keeps fission products from escaping. (SiC)
- 5) Outer Pyrolytic Carbon - Protects SiC from external stresses during manufacturing, puts SiC into compression (like IPyC) (OPyC)

2) Oxide fuel kernels (e.g., UO_2) are great at oxidizing fission products when Cl_2 is separated from U during fission. Oxidized fission products are far less mobile. We also have a lot of experience with UO_2 fuel. However, excess oxygen forms CO (by interacting with graphite), which can lead to overpressurization. Carbide fuel kernels (e.g., UC_2) is less characterized and less understood overall. It prevents CO generation, but also doesn't oxidize fission products, leaving them mobile and able to attack the SiC.

A combination is now being designed and tested (80:20 $UO_2:UC_2$) to keep enough oxygen to immobilize fission products without generating too much CO because the oxygen preferentially bonds with U from UC_2 first.

3) Fission products are often divided into five groups:

- Gases (e.g., Xe, Kr) - increase pressure and try to escape but don't interact
- Volatile (e.g., I, Cs) - vapor in hot regions but condense in colder regions.
- Metals (e.g., Ag, Pd) - Migrate or form deposits in solid form without interacting. Pd reacts with SiC to form SiPd species, degrading the pressure boundary. Ag appears to be mobile and might pass through SiC without interacting
- Soluble oxides - oxidize but stay dissolved in solid fuel solution.
- Insoluble oxides (e.g., Zr) - oxidize but collect to form deposits in a different phase

- wanted $U(-)UO_2$ FP considerations, etc.

4) Three SiC irradiation temperature regimes:

- Low T (≤ 423 K): Amorphization regime. Not enough thermal energy to repair irradiation damage, leading to highly disordered material with black spots. Lots of swelling (increases w/ irradiation)
- Medium T (423 K $\leq T \leq 1273$ K): Point defect saturated swelling regime. Point defects dominate and lead to swelling. Higher T leads to less swelling because thermal energy allows faster healing (interstitials finding vacancies), so the saturation swelling (damage/repair equilibrium) is lower.
- High T (≥ 1273 K): Unsaturated void swelling. Voids merge together and grow faster than repair can occur. This leads to swelling that increases without saturating. Frank loops and larger voids dominate; point defects insignificant contribution to swelling. Higher T can lead to more swelling but data is sparse.

- 5) Amorphous regime: Thermal conductivity severely degraded (phonons cannot propagate)

Point Defect Saturated Swelling Regime: Thermal conductivity degrades linearly with swelling because both are caused by point defects (expands crystal structure while interrupting phonon flow). Both will saturate together.

Void swelling: Voids also interrupt phonons, but not as much as point defects. Higher irradiation degrades thermal conductivity, but in a less than linear manner.

→ Point defects are the primary phonon scatterers.

- 6) Graphite exhibits a behavior called "turnaround." It shrinks with irradiation at first, but after a point, it reverses and starts to swell at increasing rates. This is due to the microstructure of nuclear graphite.

During manufacturing, the graphite cools from very high temperatures. Due to an anisotropic coefficient of thermal expansion, it has high strain at grain boundaries, leading to lenticular voids parallel to basal planes known as Mrozowski cracks.

During irradiation, interstitial atoms bond between basal planes, causing local expansion in the c-axis, and contraction in the a-axis. Vacancies also contract the a-axis. However, the net effect is a contraction because the Mrozowski cracks close from c-axis stress, masking c-axis expansion.

After the void spaces close up, the c-axis expansion is no longer masked and it dominates dimensional changes leading to a net positive swelling.

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7) • Overpressure: At high burnups, and especially for oxide fuels that cause CO formation, the internal pressure from fission product gases and CO can cause catastrophic stresses on the SiC layer, making it burst and release fission products. Mitigated with oxygen getters (UCO fuel, ZrO dispersion) and limiting burnup.

• IPyC irradiation induced cracking leading to SiC Failure:

AMPAD
If IPyC irradiation causes too high of stresses (and creep cannot relieve those stresses quickly enough), the IPyC layer can crack. This alone doesn't mean the TRISO particle has failed, but it makes it much more likely for the SiC to fail because the SiC is no longer in compression (or is in much less compression), a stress concentrator can suddenly form, and fission products have a more direct route to attack the SiC. Mitigated with proper IPyC design/manufacture and limiting irradiation.

• Kernel Coating Mechanical Interaction (KCMI):

More of a concern with historical TRISO particle geometries, KCMI occurs when the fuel kernel swells enough to mechanically interact with stiff surrounding layers, causing them to rupture. This is mitigated by designing a thick enough buffer layer to accommodate the expected expansion at max burnup (or limit burnup).

→ ZrC, ZrO₂ would be very different

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8) Replacement of SiC with ZrO is of interest because ZrO can withstand much higher temperatures without degrading. It can enable higher operating temperatures and improve safety in certain accident conditions. It is also resistant to Pd attack, closing a vulnerability of SiC. However, it is not oxidation resistant, leading to manufacturing challenges and vulnerability during accident conditions where water or oxygen are introduced. We don't have as much experience or data on ZrO, but it remains of interest after SiC is fully qualified.

9) The IHX must withstand coolant temperatures, erosion (coolant is traveling very fast), and chemical attack from impurities (He itself is not corrosive). Alloy 617 is the most suitable candidate, though it is not qualified for the full temperature range desired.

Impurities even in trace amounts (tens of ppm) can cause excessive corrosion, carburization, or decarburization. Water or O_2 can oxidize the surface corrosively. Carbon-containing species like CO , CO_2 , or CH_4 can carburize the surface, leading to excessive carbon precipitates causing low temperature embrittlement.

Too low of carbon activity can cause the opposite - decarburization, where C deposits leech out and the material loses creep resistance.

A passivated oxide on the surface (e.g., Cr_2O_3) is needed to lock in the right carbon content and prevent chemical corrosion like unstable oxide layers.